

Geometry-native primitive replacement

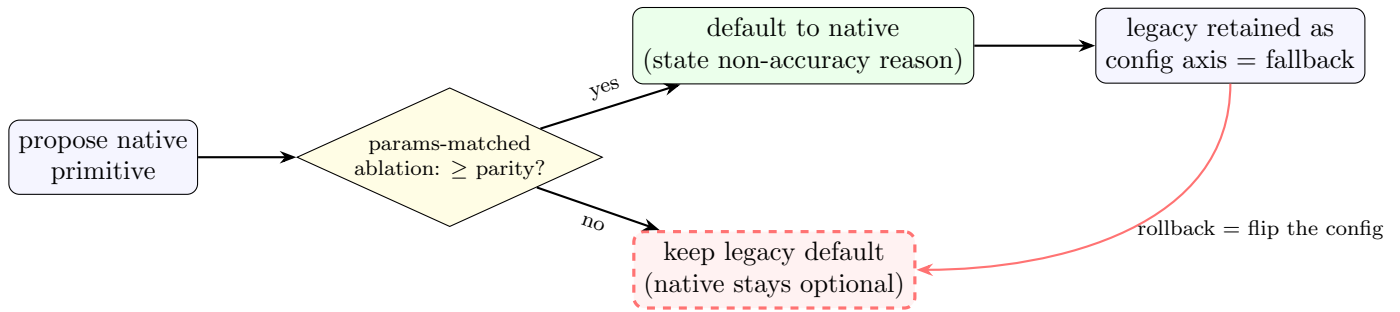
A parity-gated design principle (measurement-grounded, fallback-mandatory)

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The principle. Across the framework, prefer *learnable, geometry-/structure-native* primitives over fixed black-box ones — `tanh` → Catmull-Rom (CR) spline, plain aggregation → signed rotor propagation, node-ID table → signed structural features, `softmax` → rotor scoring. **But** (the two binding conditions): a native primitive becomes the default *only* after a parameter-matched ablation shows $\geq$ **parity by measurement**, and the legacy primitive is **always retained as a config-axis fallback**. The justification for the swap is *consistency / boundedness / quantization / hardware fit / auditability* — **not** a claimed accuracy gain, because the fair ablations to date show parity (and one readout case where the simple op wins).

The decision gate (every swap)



The ledger (what is measured, what is still a bet)

swap	status		evidence / fallback
edge activation tanh →CR spline	<i>parity</i> on accuracy	MEASURED	toy non-monotone probe, 3-seed: cr≈tanh at matched params; a wider tanh edges ahead <i>per param</i> ⇒ no accuracy buy. RL test = the campaign. fallback activation∈{cr,tanh}
propagation → signed rotor slerp/nlerp	first confirmed <i>lift</i>	MEASURED	+0.0045 alpha / +0.0105 etc, 5-seed, leakage-clean (2026-06-17-signed-rotor-slerp) fallback propagation off
embedding geometry: rotor S ³ vs MLP-embed	<i>parity</i> , not load-bearing	MEASURED	±0.002–0.005, inside σ (2026-06-18-rotor-vs-mlp-embe) fallback MLP-embed
readout head geometric vs real bilinear	legacy <i>wins</i> (boundary!)	MEASURED	complex/quaternion/geodesic <i>wors</i> (2026-06-17-link-head-ablatio) do not geometric-ise the output
attention softmax → rotor scoring	parity-likely; benefit off-manifold	HYPOTHESIS	rotor ”doesn’t kill performance” generalised from above; fallback score∈{softmax,rotor}, gate first
control backbone MLP→HSiKAN	parity so far	MEASURED/HYPOTHESIS	cart-pole + Galambos single-seed tie; multi-seed in flight; fallback kind∈{mlp,hsikan}
rotor output → int8 quantization (Nagare)	calibration-free quantize	HYPOTHESIS	bounded S ³ output; the <i>discriminator</i> is planned, not r (2026-06-18-rotor-embedded-na

The rules (binding)

1. **Measured parity gate.** No native primitive becomes the default without a params-matched ablation showing $\geq$ parity (median gap within cross-seed IQR). Until then it ships *off by default*, available behind its config axis. “Rotor didn’t hurt once” is a datapoint, not a licence.
2. **Mandatory fallback.** Every swap is a *config axis* that retains the legacy primitive (activation∈{cr,tanh}, score∈{softmax,rotor}, kind∈{mlp,hsikan}). Rollback is flipping the flag — no code change, no migration.
3. **Honest justification.** State the reason as consistency / boundedness / quantization / hardware / auditability. Do *not* claim accuracy; the fair ablations show parity, and the *readout* is a measured counter-example where the simple op wins — a hard boundary of the principle.
4. **Distinguish in writing** what is measured from what is a hypothesis (this ledger), and run the discriminating ablation before promoting a hypothesis row to a default.

Status note: the CR activation swap (hymeko.rl/policy.py) + its activation∈{cr,tanh} fallback are implemented and **verified** (15 policy tests incl. a parity test vs signedkan.wip). The toy A/B above shows **accuracy parity**, so CR ships as the default on the *consistency/KAN-structure* rationale — not an accuracy claim — exactly per this principle. The real-topology RL read is the in-flight CR campaign.